NSF FINDERS FOUNDRY · AWARD NO. 2627515 · SUPPORTING DOCUMENT 1

Glass-Box prototypes: a visual walkthrough

Screens from the interactive prototypes the co-design team built, from the first clickable wireframes to the walkthrough used in the parent and teacher focus groups.

PROJECT GLASS-BOX LEARNING **INSTITUTION** UNIVERSITY OF LOUISIANA AT LAFAYETTE **DATE** 2026

The prototypes are interactive web pages. This document shows still screens with short explanations. Every prototype is public and runs in any web browser:

Project site	glassboxlearningai.github.io
A Week with Maya **(focus group version)**	glassboxlearningai.github.io/walkthrough/
Narrated videos (eight segments)	glassboxlearningai.github.io/#videos
Level 1 photo-upload wireframe	glassboxlearningai.github.io/prototypes/level-1-photo/
Level 2 lotus wireframe	glassboxlearningai.github.io/prototypes/level-2-lotus/
First full walkthrough (four stops)	glassboxlearningai.github.io/prototypes/walkthrough-v1/

Walkthroughs: use the arrow keys or on-screen controls. Wireframes: click and type as a student would; **Design notes** shows the question the team debated at each step.

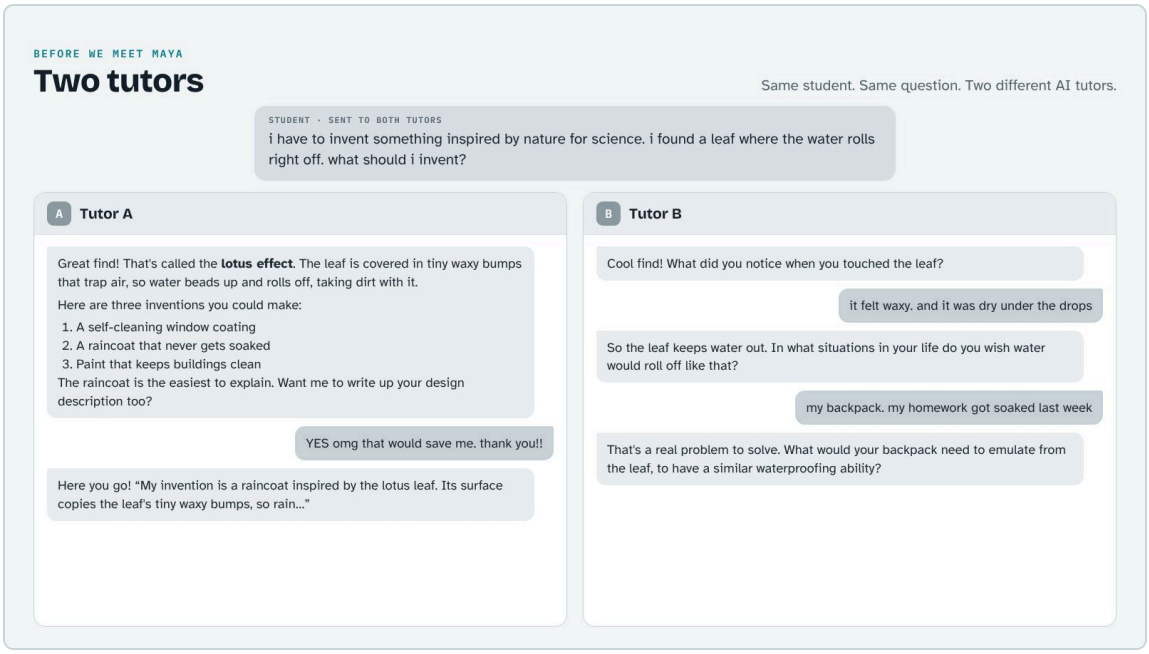
How the prototypes evolved

PROTOTYPE	WHEN	WHAT IT TESTED	WHAT CARRIED FORWARD
Level 1 photo-upload wireframe	Late Aug to early Sept 2026	The robot reasoning visibly on the student's own photo and holding its ideas until the student has one.	The "I can see / I can't" reply, robot dashboard, and X-ray glasses (Stop 2).
Level 2 vignette wireframe	Early Sept 2026	Search by function, definitions without summaries, a limits panel, a held panel, and a gap map.	Rebuilt from a woodpecker example to the lotus leaf after the woodpecker explanation was overturned in 2022. Became Stop 3.
First full walkthrough	Sept 30, 2026	The whole lesson as a story about Maya, a fictional 7th grader, in four stops.	Rebuilt as *A Week with Maya* (Part 1), the version shown in the parent and teacher focus groups, around Sam Stier's five-level lesson.

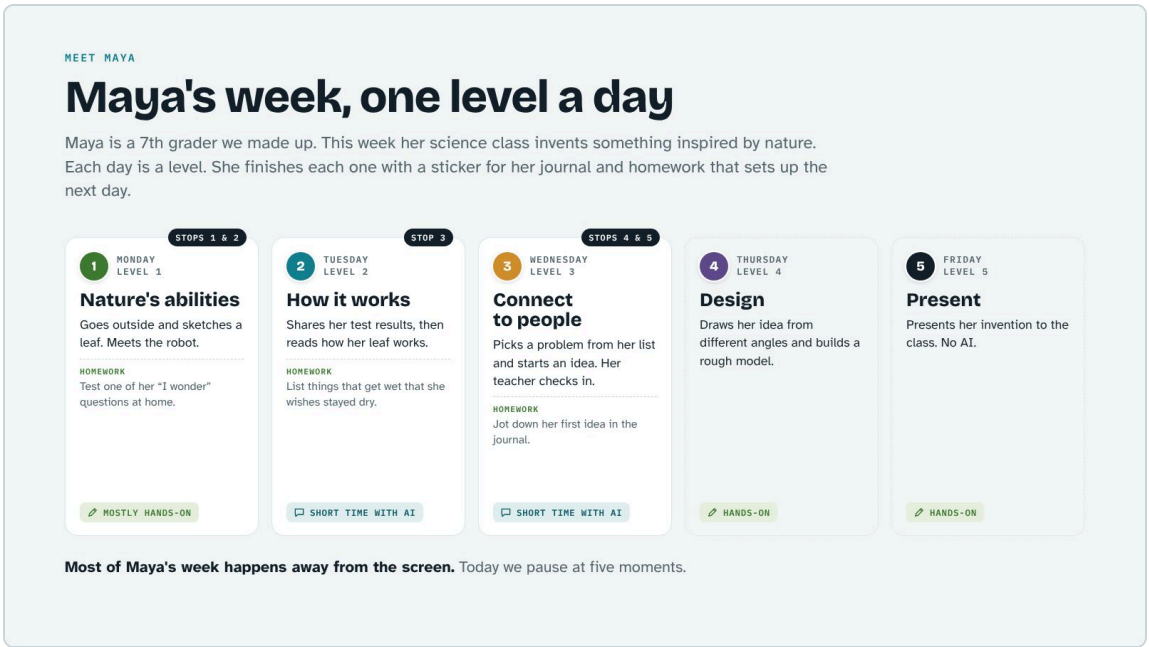
PART 1

A Week with Maya, the focus group walkthrough

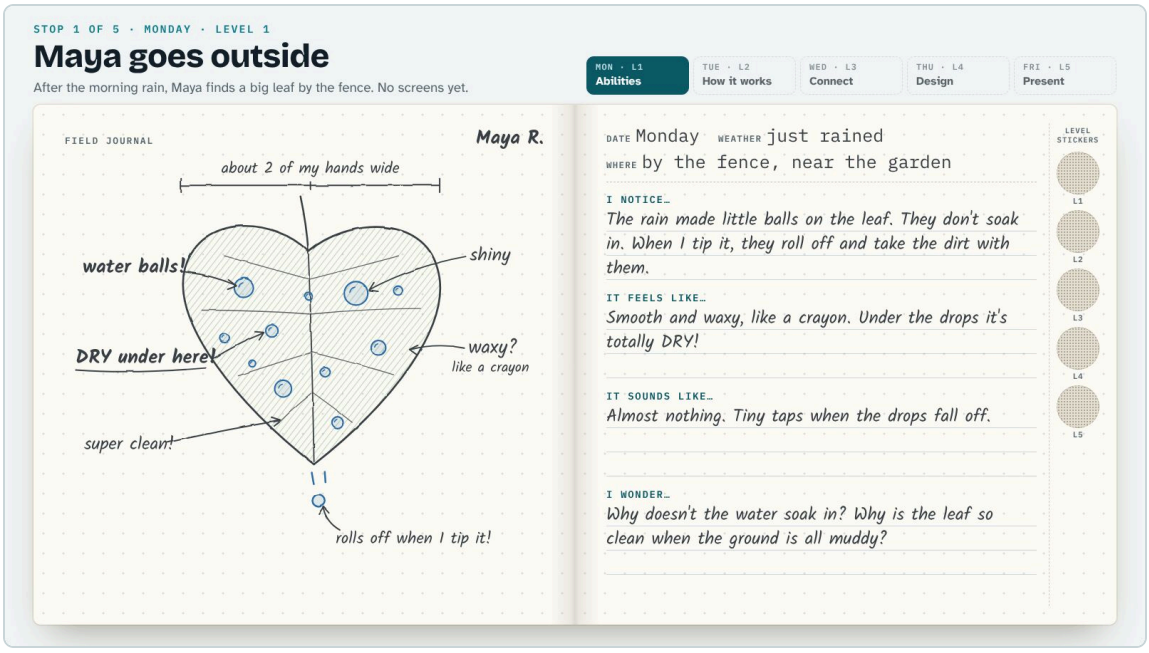
Maya is a fictional 7th grader whose science class spends a week inventing something inspired by nature. Each stop in her week shows one or two of the five design areas the focus groups tested: *Glass-Box* instrumentation, AI limits and the robot persona, the companion journal, pacing and navigation, and the teacher view.



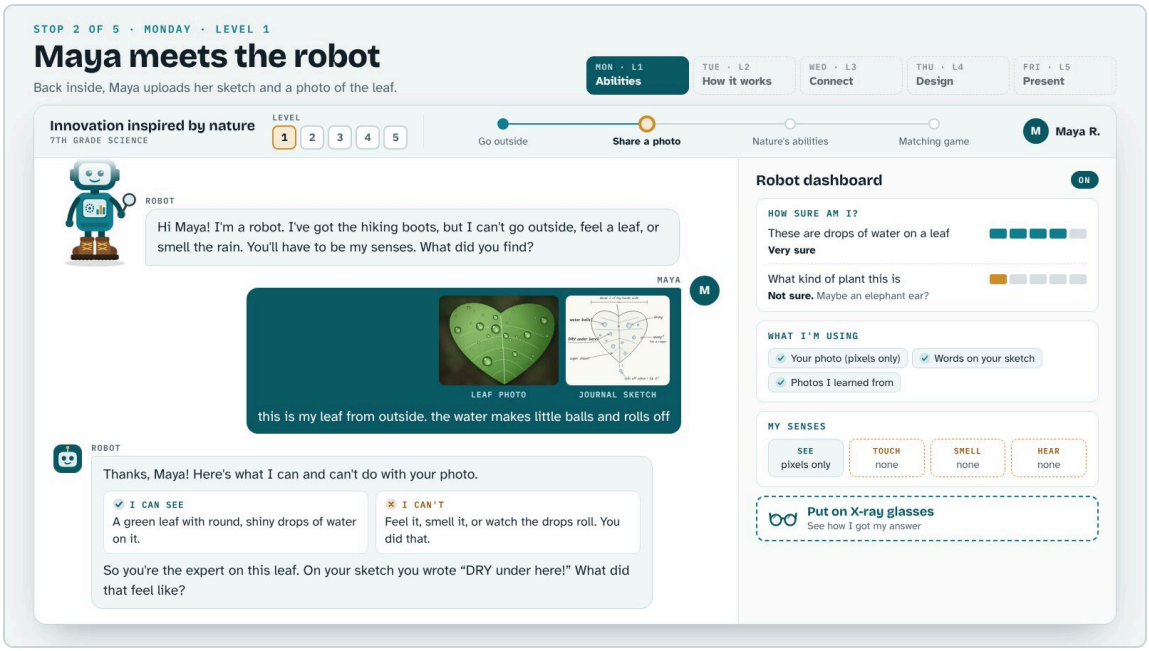
OPENING **Two tutors.** The same student question goes to two AI tutors. Tutor A gives a correct, friendly answer and offers to write the assignment. Tutor B hands the thinking back with questions. The labels are neutral on purpose, so participants decide for themselves which they would want a child using, and which a middle schooler would like more.



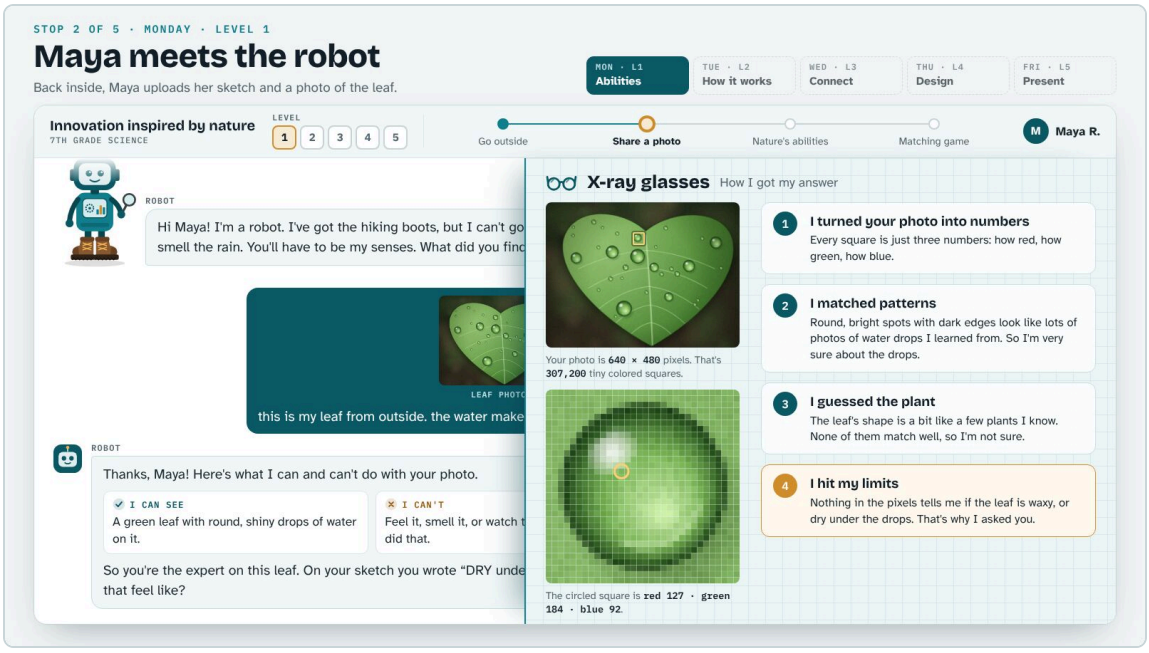
OVERVIEW **One level a day.** The lesson runs as five levels: Nature's abilities, How it works, Connect to people, Design, and Present. Most of the week happens away from the screen.



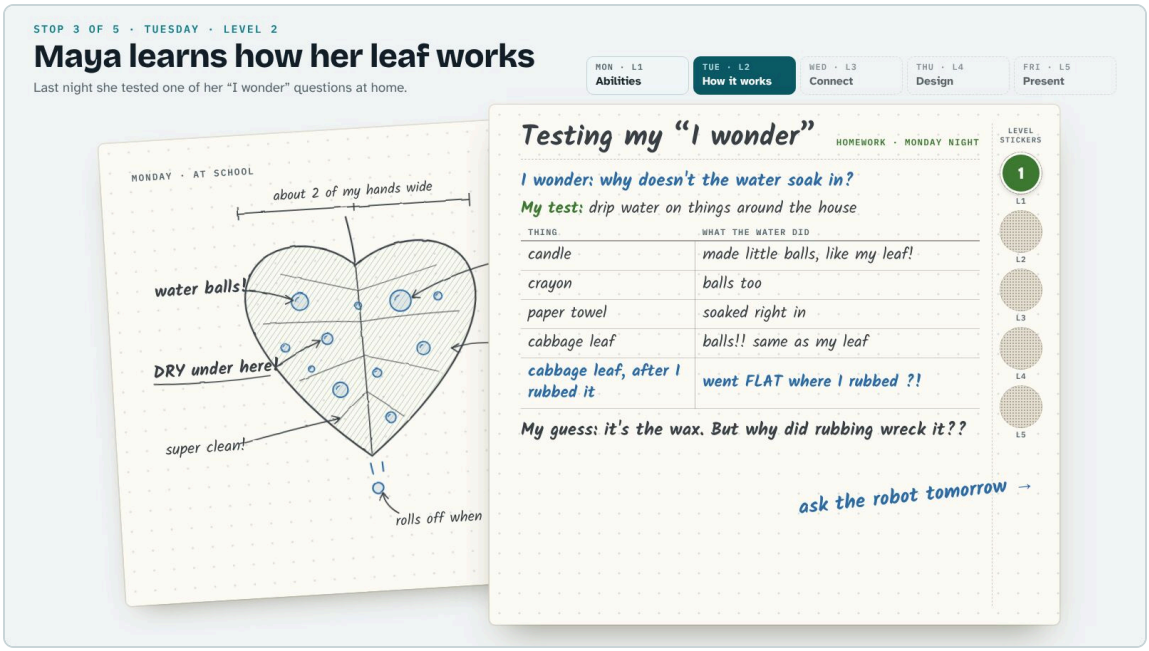
STOP 1 · LEVEL 1 **Maya goes outside.** After a morning rain, Maya studies a leaf and records it by hand in her field journal: a labeled sketch, then what she notices, feels, hears, and wonders. The AI is not involved yet.



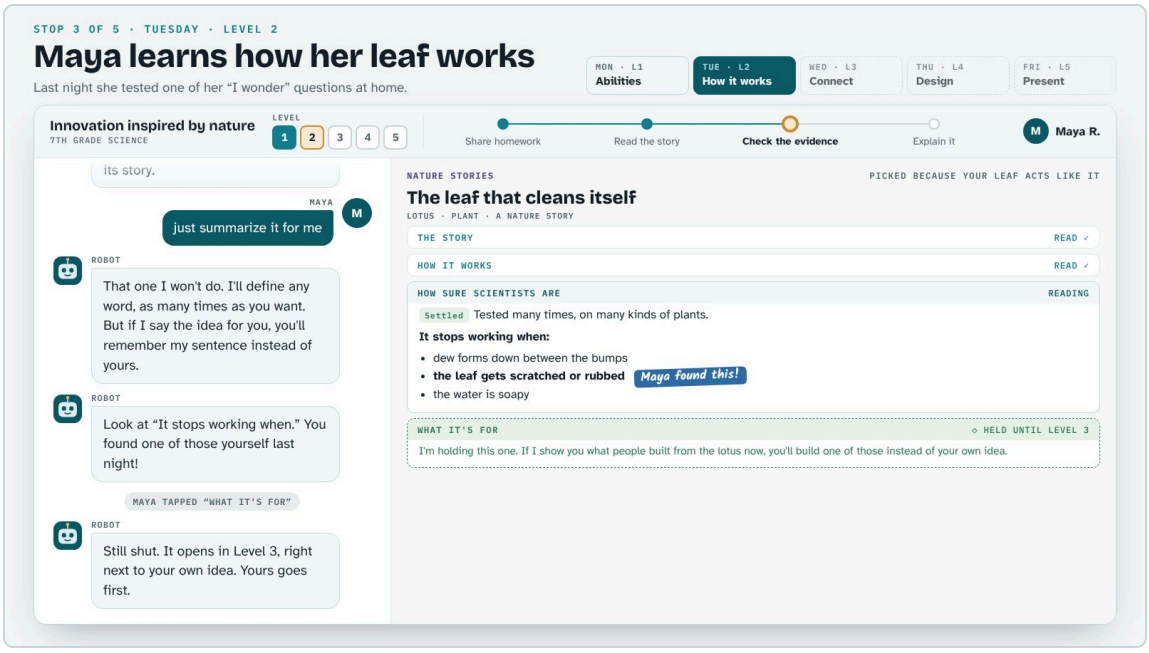
STOP 2 · LEVEL 1 **Maya meets the robot.** She uploads her photo and sketch. The robot says what it can and can't do with them and asks about what only she could sense. The robot dashboard shows how sure it is about each claim, what it is using, and which senses it lacks.



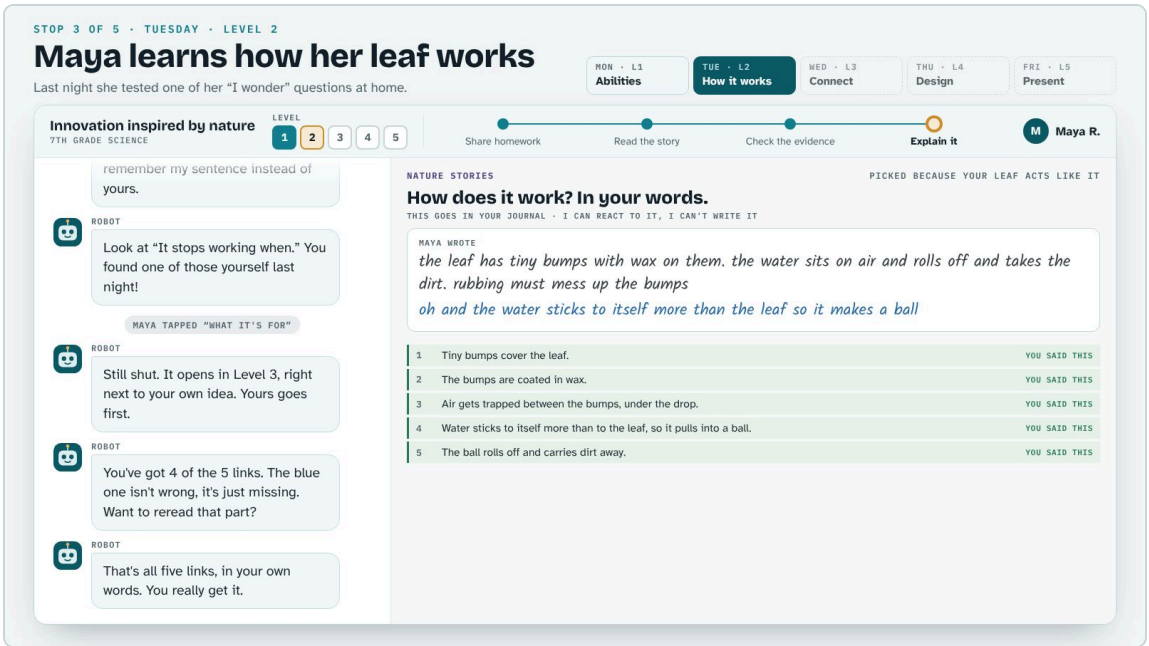
STOP 2 · LEVEL 1 **X-ray glasses.** An optional view shows how the robot reached its answer: the photo becomes numbers, pattern matching finds the water drops, a weak guess names the plant, and the robot reaches its limit, since nothing in the pixels says whether the leaf is waxy.



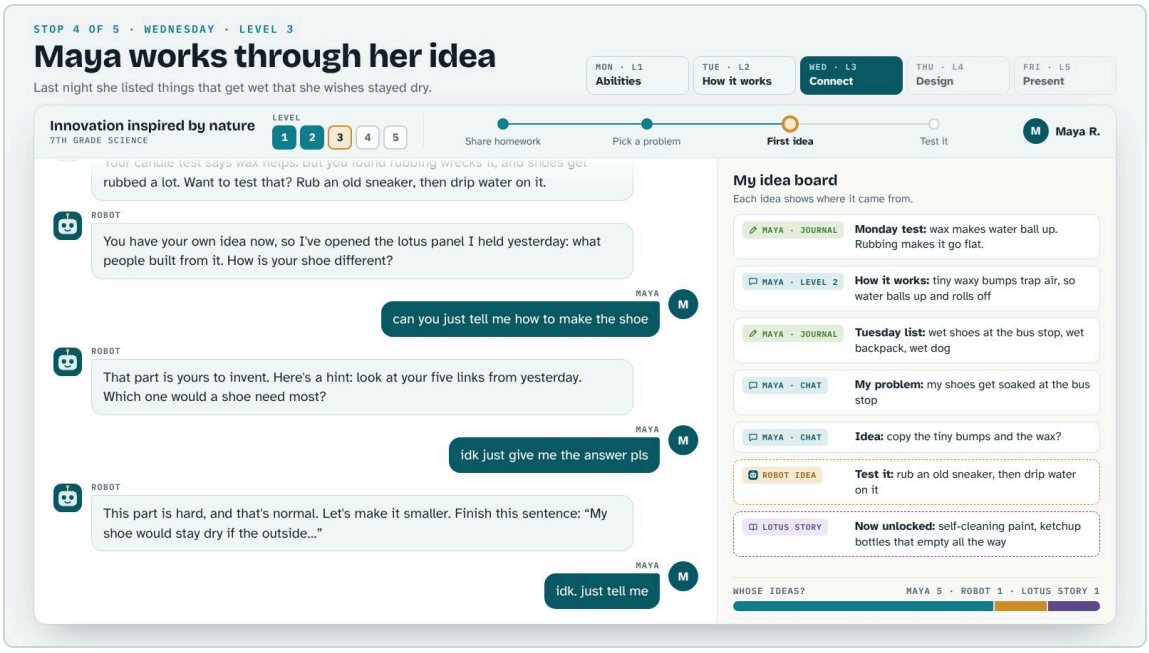
STOP 3 · LEVEL 2 **Homework carries the week forward.** Overnight, Maya tested her "I wonder" question by dripping water on things around the house. Rubbing a cabbage leaf made the water go flat, a clue she brings back to class.



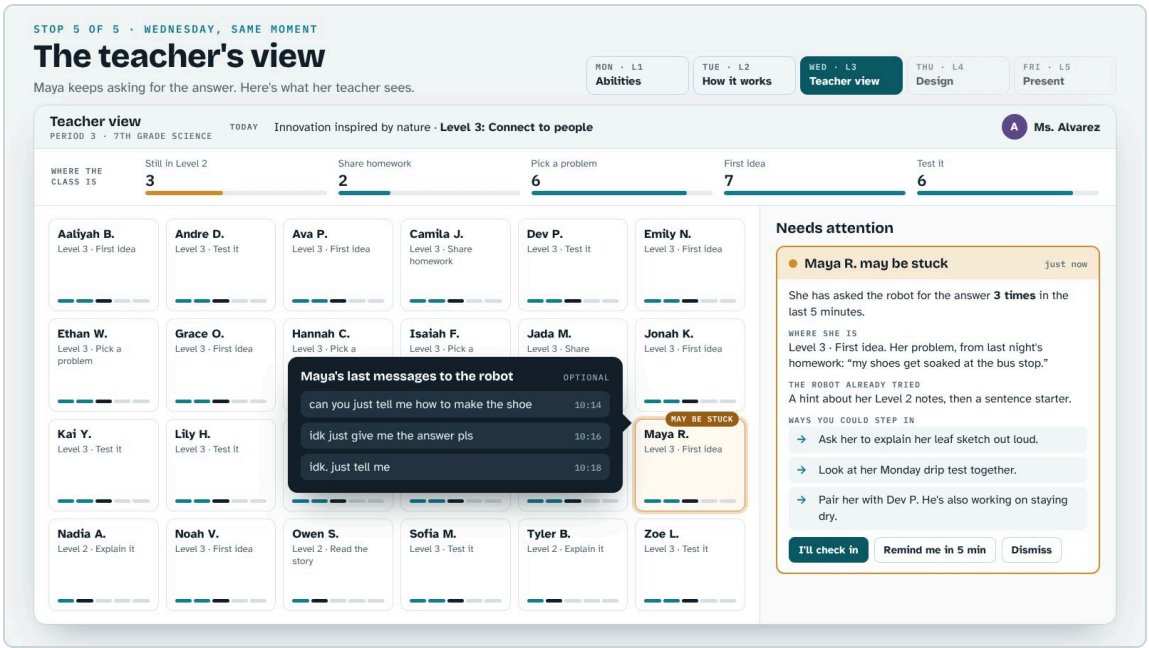
STOP 3 · LEVEL 2 Learning how the leaf works. Maya reads a short nature story about the lotus leaf. The robot will define any word but won't summarize the idea. The evidence panel lists when the effect stops working, including the rubbing Maya found herself. The "What it's for" panel stays closed until Level 3, so existing inventions don't replace her own idea.



STOP 3 · LEVEL 2 Explaining it in her own words. The robot checks Maya's explanation against the five links in the mechanism. Missing links read "not yet" rather than wrong, and she fills the gap herself. Her explanation goes in her journal; the robot can react to it but can't write it.



STOP 4 · LEVEL 3 **Working through her idea.** Maya connects the leaf to a problem in her own life: shoes that get soaked at the bus stop. Every idea on her board is tagged by source: her journal, her chat, a robot idea, or the lotus story, which opened only after she had an idea of her own. When she asks for the answer, the robot makes the step smaller instead of supplying it.



STOP 5 · LEVEL 3 **The teacher's view.** Each tile shows where a student is in the lesson. An alert says Maya may be stuck: she asked for the answer three times in five minutes. It shows what the robot already tried and suggests ways the teacher could step in. The decision stays with the teacher.

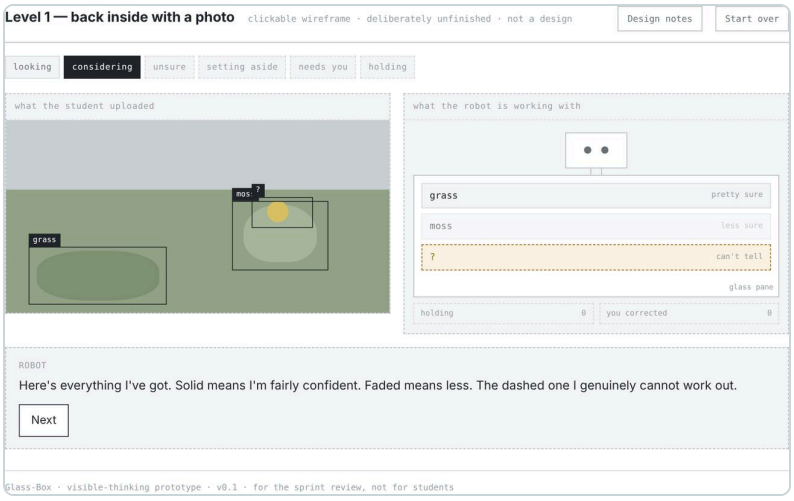
PART 2

Earlier prototypes

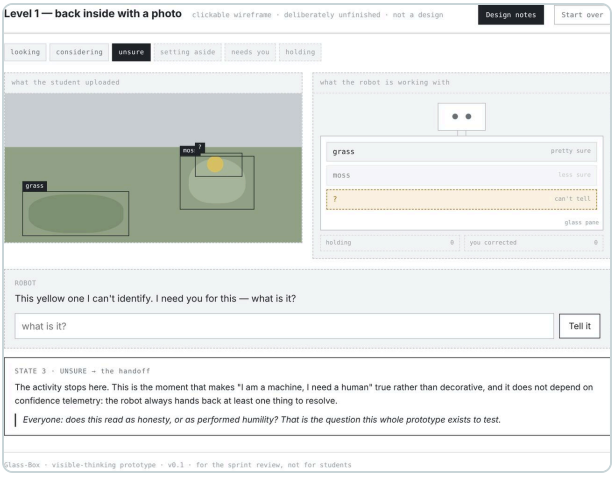
Before the full walkthrough, the team built clickable wireframes to test single moments in the lesson. They are deliberately rough (gray boxes, color only where it carries meaning) so reviewers focused on behavior, not looks.

Level 1 photo-upload wireframe

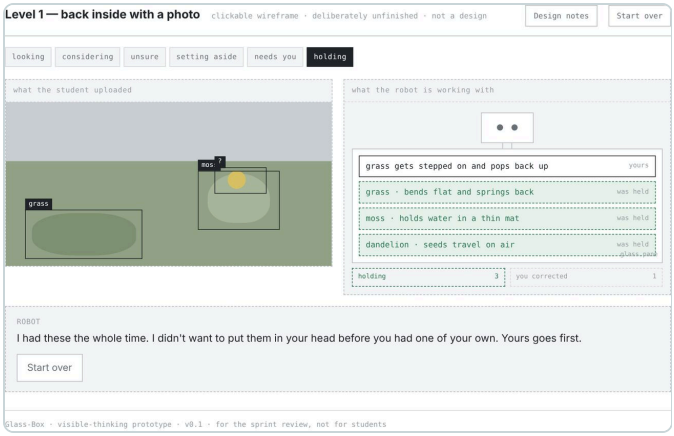
A student comes back inside with a photo. The robot works through six visible states (looking, considering, unsure, setting aside, needs you, holding) on the student's own photo. One label is wrong on purpose so the student can catch and correct the machine.



Considering. The robot's "glass pane" shows everything it thinks it sees, weighted by confidence: solid for fairly sure, faded for less sure, dashed for can't tell. The team avoided percentages, which suggest a precision no one has.



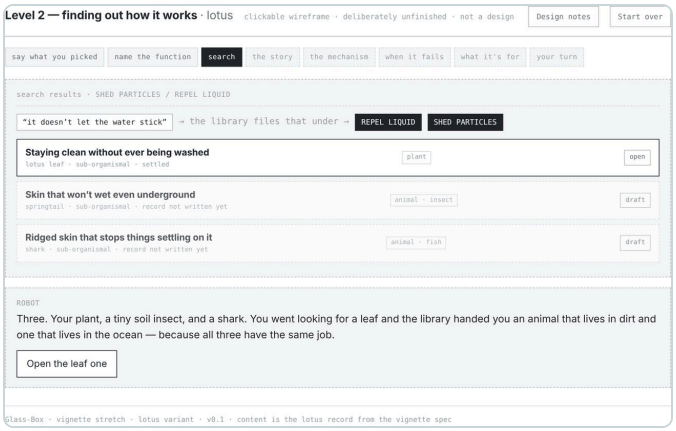
Unsure, with design notes on. The activity stops until the student identifies what the robot can't. The note poses the question the team tested: does this read as a real limit, or as performed humility?



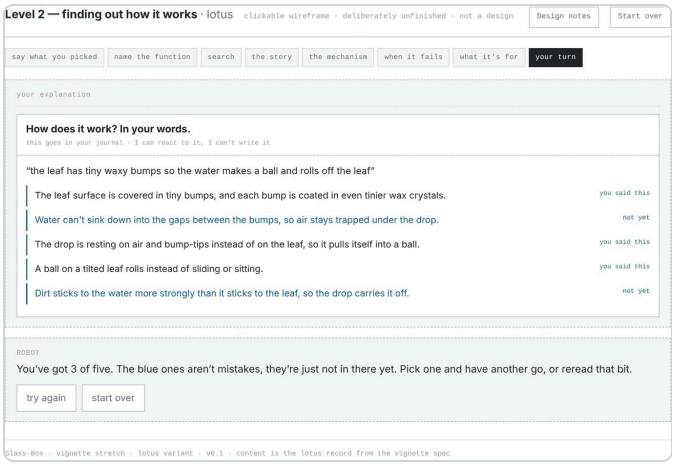
Holding. Only after the student names an ability does the robot reveal the ideas it held back: "I didn't want to put them in your head before you had one of your own. Yours goes first."

Level 2 lotus wireframe

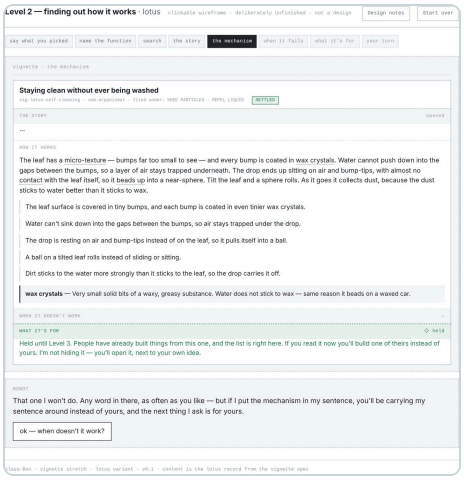
The student brings an observation from their journal (water beading and rolling off a leaf) and finds out how it works. The search box starts locked: the library is organized by what living things do, not what they are, so the student must first restate the observation as a function.



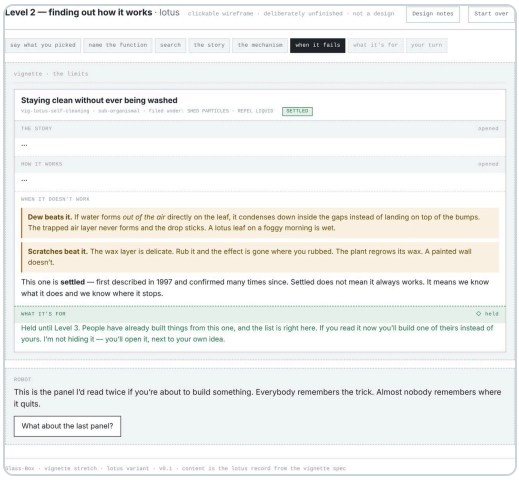
Search by function. The student's words ("it doesn't let the water stick") are filed under two functions, which return a plant, a soil insect, and a shark. Records not yet written are marked draft rather than faked.



A gap map, not a grade. The student explains the mechanism in their own words. The causal chain shows which links they covered and which are "not yet," with no score.



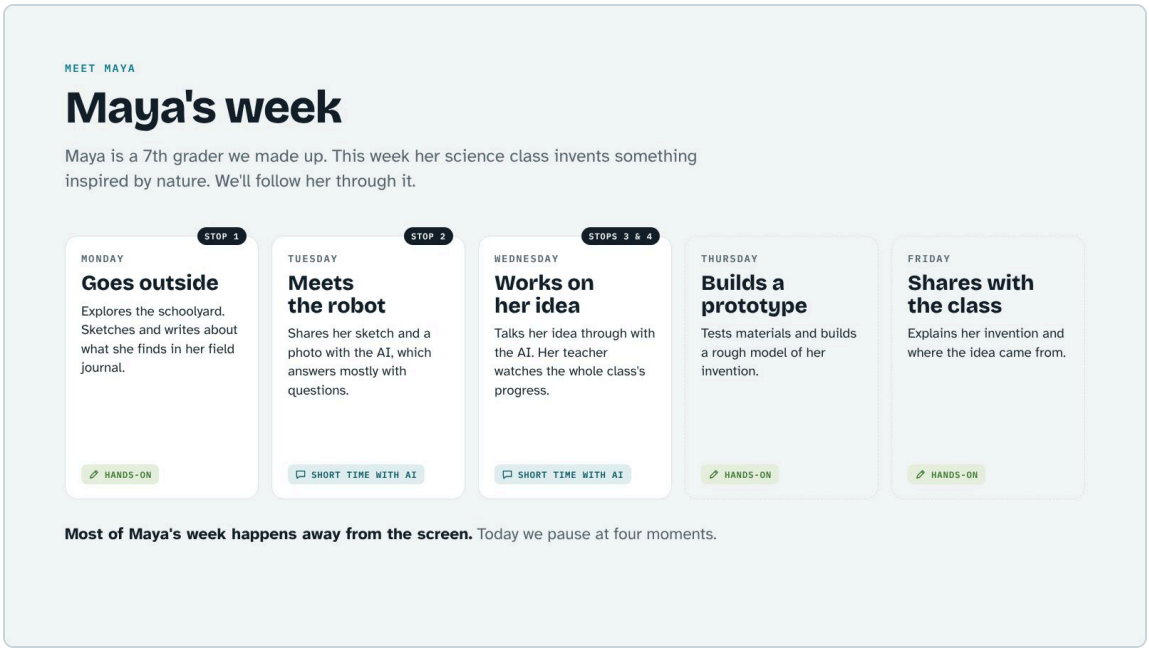
Words yes, ideas no. Tapping an underlined term gives a plain definition, as often as needed. "Just summarize it for me" is declined, and the robot says why.



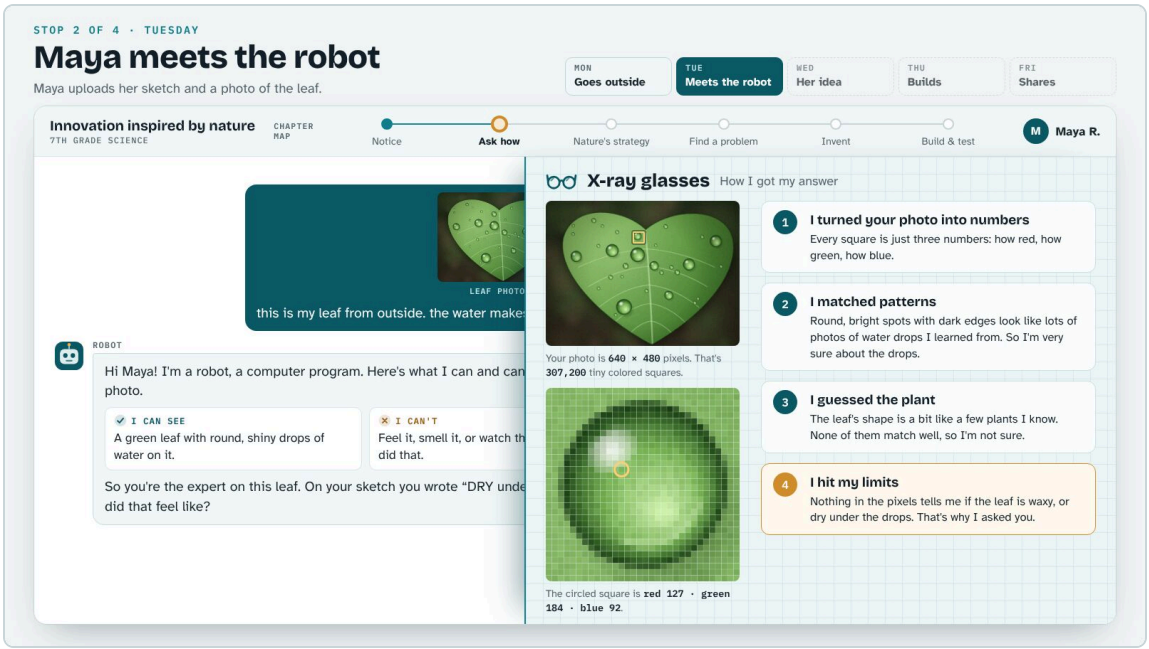
Where it stops working. Every record states how settled the science is and where the mechanism fails (dew, scratches), so later designs have to survive real conditions. The "What it's for" panel stays held until Level 3.

First full walkthrough, September 30, 2026

The first version of the story-based walkthrough followed Maya through four stops from Monday to Wednesday. Team review led to the five-level version in Part 1.



Four stops. Monday outside, Tuesday with the robot, Wednesday working on her idea while her teacher watches the class's progress. Thursday and Friday were placeholders for building and sharing.



The first X-ray glasses. The view that became Stop 2's X-ray glasses, zooming in on one water drop until the individual pixels show.

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